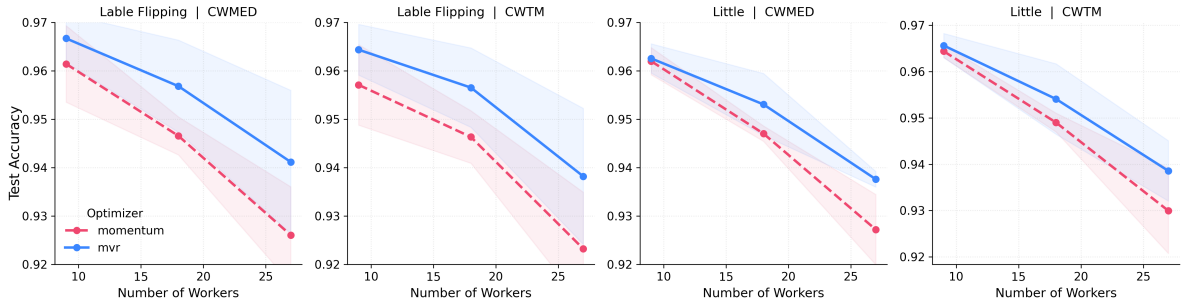


Figure 1: Test accuracy vs. number of **workers** on MNIST for synchronous learning with Byzantine fraction $\delta = 1/3$. The gap between MVR and momentum grows with the number of workers (consistent with our theoretical results).

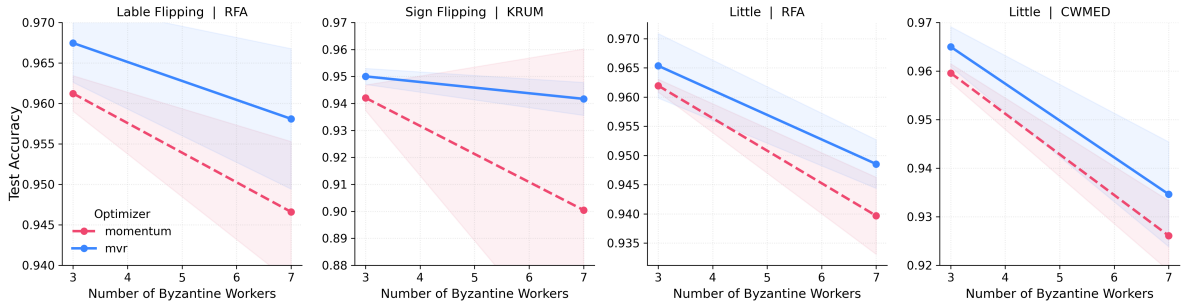
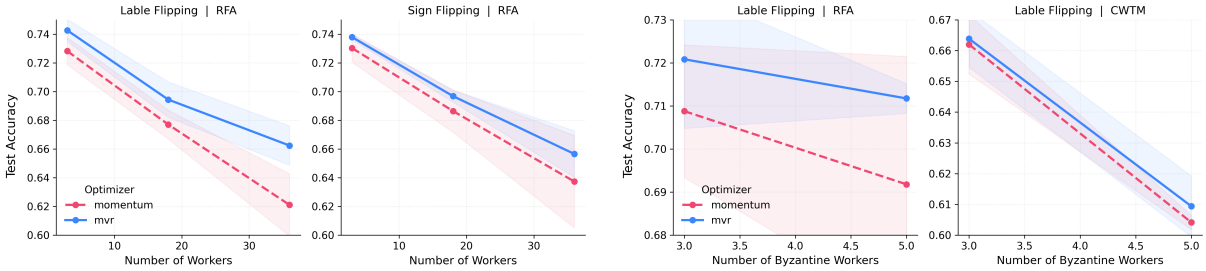


Figure 2: Test accuracy vs. number of **Byzantine workers** on MNIST for synchronous learning with 17 workers. The gap between MVR and momentum grows with the fraction of Byzantine workers (consistent with our theoretical results).



(a) Over **workers** with Byzantine fraction $\delta = 1/3$.

(b) Over **Byzantine workers** with 17 workers.

Figure 3: Test accuracy on CIFAR-10 under synchronous training. The gap between MVR and momentum increases with the number of workers/Byzantine fraction (consistent with our theoretical results).

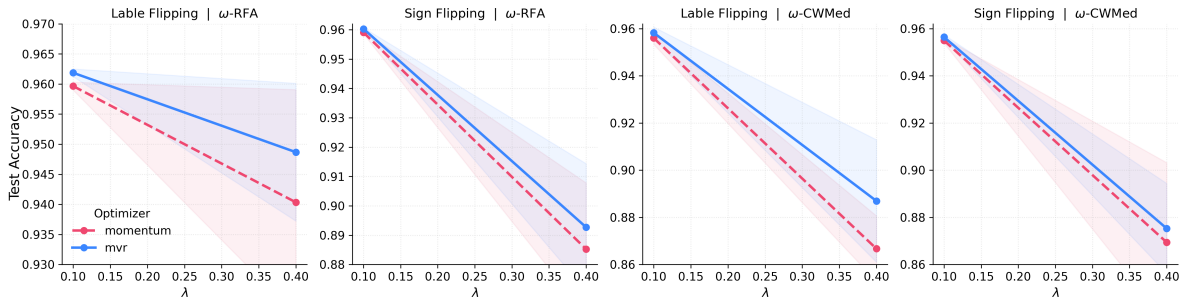
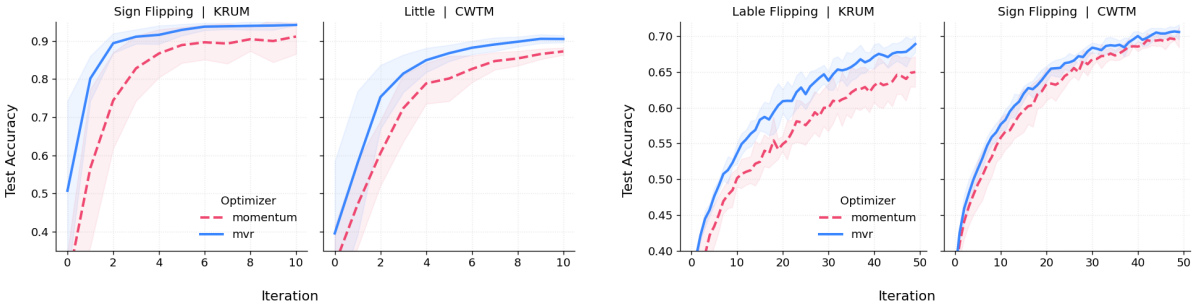


Figure 4: Test accuracy vs. the fraction of **Byzantine iterations** on MNIST for asynchronous learning with 17 workers, including a very fast Byzantine worker. The gap between MVR and momentum grows as the fraction of Byzantine iterations increases (consistent with our theoretical results).



(a) MNIST. 8/17 Byzantine workers.

(b) CIFAR-10. 4/17 Byzantine workers.

Figure 5: Test accuracy under synchronous training with additional aggregators (KRUM and CWTM).